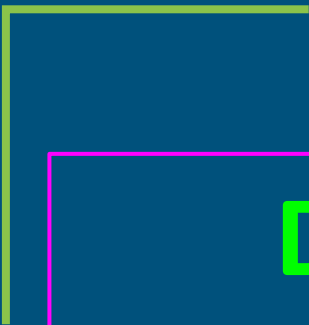
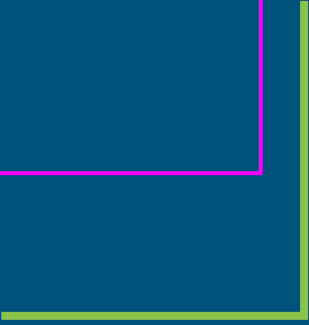




Database Design & Web Implementation CSCI-UA060



Database Fundamentals



Basic Components of a Database

- Tables
- Columns
- Rows
- Relationships

Tables

- Sometimes called entities or relations
- When analyzing a business, think of them as the nouns or things in the business
- In a library, some examples of tables might be book, author or customer
- Tables are like Excel spreadsheets with columns and rows

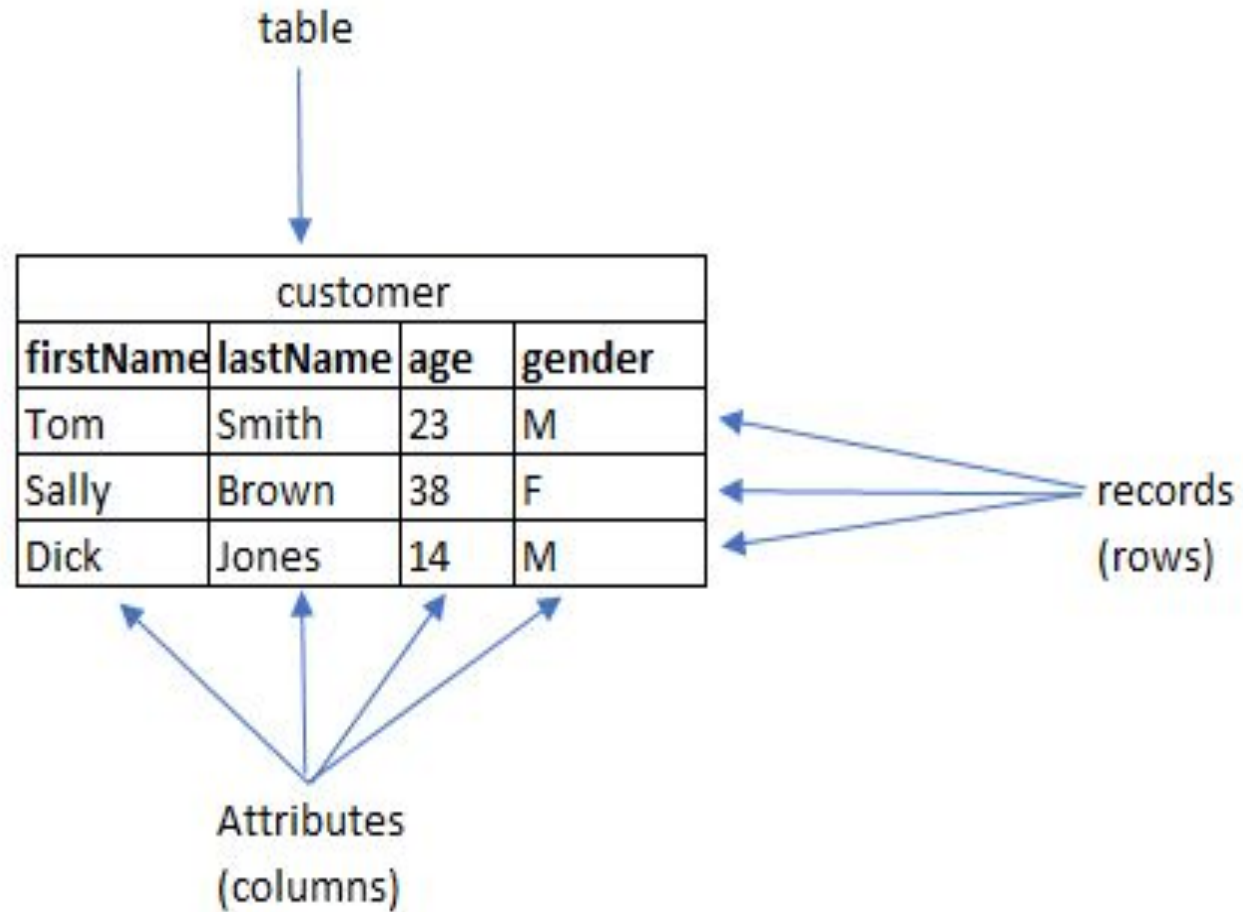
Columns

- Sometimes called attributes or fields
- Attributes describe the nouns (entities)
- In a library, a customer might have a name and an age

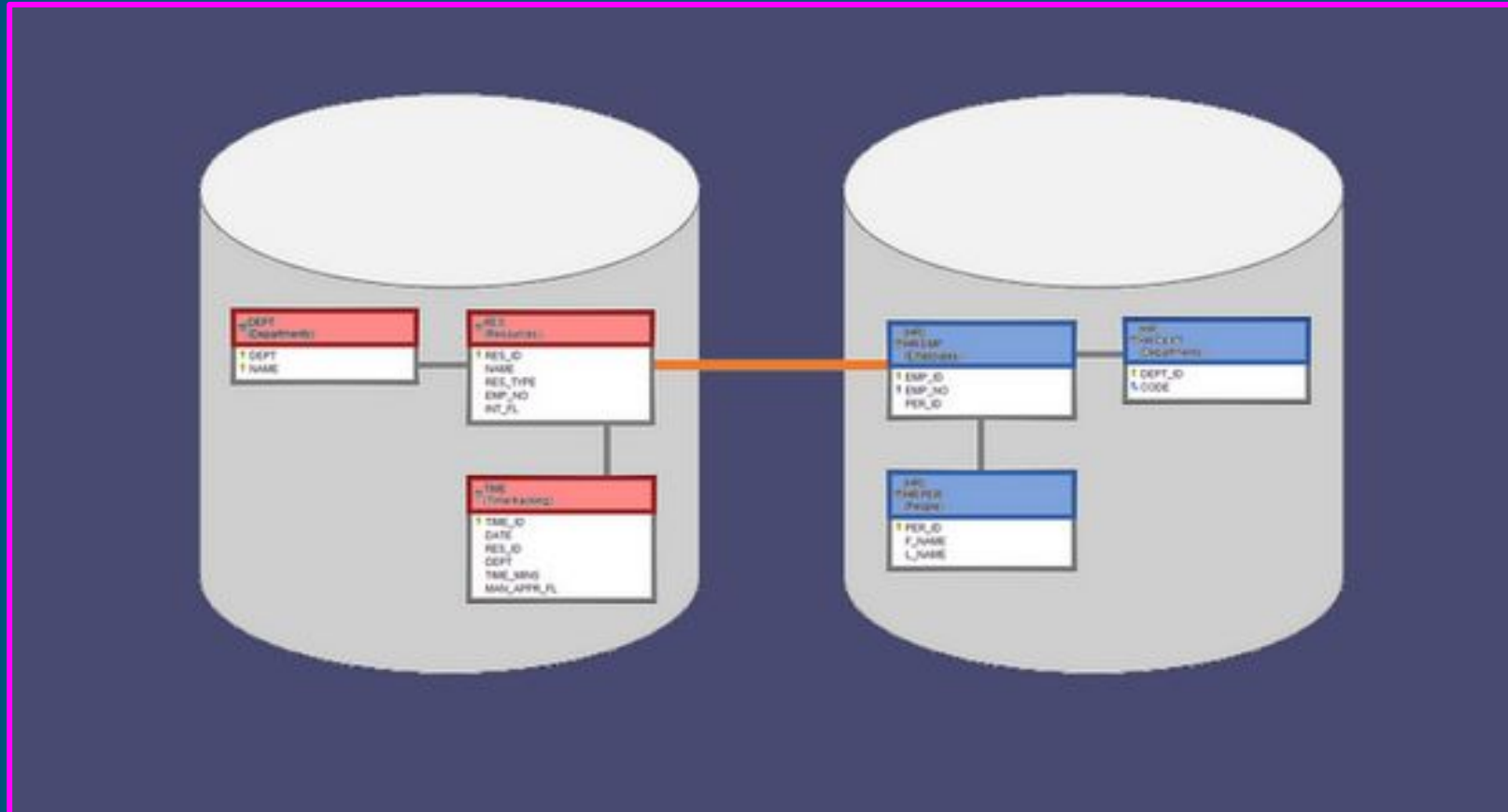
Rows

- Sometimes called records or tuples
- Each row contains the data for one instance of an entity
- In the library customer table, there will be one row for each customer, with the customer's name and age, etc

How it comes together



A collection of tables is a database



Let's draft a speculative table together in lucid app

- Possible Entities: library, students and classes, streaming services, campus dining/orders, etc.
- What attributes would we need ? What are the nouns in this system?
- What did we just define?



**When would we need another
database?**

❑ The purpose is different

- ❑ Library database vs Payroll database
- ❑ Student records vs Course catalog
- ❑ Different goals, different system

❑ The users are different

- ❑ Internal admin data vs public-facing data
- ❑ Professors vs students

- ❑ The data lifecycle is different

- ❑ Transactional data vs archived data
 - ❑ Live app data vs analytics

- ❑ The system needs isolation

- ❑ Security
 - ❑ Performance
 - ❑ Legal/privacy boundaries